Neuropsychology & Biology

Comprehensive EPPP Preparation Study Guide
For Destynee | *Dr. DeDe's Concept List*

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1. Down Syndrome

Definition: Genetic condition caused by trisomy 21 (three copies of chromosome 21).

Characteristics: - Intellectual disability (typically mild to moderate; average IQ ~50) - Hypotonia (low muscle tone) - Distinctive facial features (upward slanting eyes, flat facial profile) - Increased risk of congenital heart defects, leukemia, hypothyroidism, Alzheimer's disease - Higher risk of hearing/vision problems, sleep apnea

Life Expectancy: 60+ years with appropriate care.

Source: DSM-5-TR (2022); Bull, M. J. (2022). Health supervision for children with Down syndrome. *Pediatrics*.

2. Angelman Syndrome

Definition: Genetic disorder caused by deletion or inactivation of genes on the **maternally-derived chromosome 15**.

Characteristics: - Severe intellectual disability (IQ typically <50) - Minimal or absent speech - Ataxia (unsteady gait, tremulous movements) - Frequent laughter/smiling, excitable personality - Seizures, microcephaly - Happy, affectionate demeanor - Hand flapping, short attention span

Genetic Imprinting Disorder: Contrast with Prader-Willi (paternal deletion).

Source: Williams, C. A., et al. (2006). Angelman syndrome 2005: Updated consensus. *American Journal of Medical Genetics Part A*.

3. Prader-Willi Syndrome

Definition: Genetic disorder caused by deletion or inactivation of genes on the **paternally-derived chromosome 15**.

Characteristics: - Infantile hypotonia ("floppy baby") and feeding difficulties - Failure to thrive in infancy, hyperphagia after age 2 - Food obsession, obesity if not controlled - Mild to moderate intellectual disability - Hypogonadism, short stature, small hands/feet - Behavioral problems (temper tantrums, obsessive-compulsive features, stubbornness) - Distinctive facial features (almond-shaped eyes, narrow forehead)

Genetic Imprinting Disorder: Contrast with Angelman (maternal deletion).

Source: Cassidy, S. B., et al. (2012). Prader-Willi syndrome. *Genetics in Medicine*.

4. Fragile X Syndrome

Definition: Most common inherited cause of intellectual disability; caused by trinucleotide repeat expansion (CGG) on X chromosome (FMR1 gene).

Characteristics: - Intellectual disability (mild to severe; males more affected due to X-linked) - Long face, large ears, prominent jaw, high palate - Hyperactivity, social anxiety, gaze aversion - Autism spectrum features common (~30% meet criteria) - Connective tissue problems (flexible joints, flat feet)

Genetic Pattern: X-linked; females can be carriers with milder symptoms.

Source: Hagerman, R. J., et al. (2009). Fragile X syndrome. *Nature Reviews Disease Primers*.

5. Williams Syndrome

Definition: Rare genetic disorder caused by microdeletion on chromosome 7 (7q11.23).

Characteristics: - "Cocktail party" personality (overly friendly, socially uninhibited, empathetic) - Relative strength in language and face processing - Severe weakness in visuospatial processing - Cardiovascular problems (supravalvular aortic stenosis common) - Intellectual disability (mild to moderate) - Distinctive facial features (broad forehead, short nose, full cheeks) - Musical abilities often spared

Cognitive Profile: Uneven abilities with stark contrast between language/face processing vs. spatial reasoning.

Source: Mervis, C. B., & Klein-Tasman, B. P. (2000). Williams syndrome. *MRDD Research Reviews*.

6. Klinefelter's Syndrome

Definition: Genetic condition (XXY karyotype) affecting males.

Characteristics: - Tall stature with long legs - Small testes, infertility, reduced testosterone - Gynecomastia (breast development) - Language and learning difficulties - Reduced facial/body hair - Social difficulties, anxiety, depression risk

Prevalence: Approximately 1 in 600-1000 male births.

Treatment: Testosterone replacement therapy.

Source: Bojesen, A., & Gravholt, C. H. (2007). Klinefelter syndrome in clinical practice. *Nature Clinical Practice Urology*.

7. Brain Structures Overview

Frontal Lobe: Executive functions, planning, decision-making, motor cortex, Broca's area (speech production).

Parietal Lobe: Somatosensory processing, spatial awareness, integration of sensory information.

Temporal Lobe: Auditory processing, Wernicke's area (language comprehension), memory (hippocampus), emotion (amygdala).

Occipital Lobe: Visual processing.

Cerebellum: Coordination, balance, procedural learning.

Brainstem: Vital functions (breathing, heart rate), arousal, sleep-wake cycles.

Source: Kolb, B., & Whishaw, I. Q. (2015). *Fundamentals of human neuropsychology* (7th ed.). Worth Publishers.

8. Amygdala

Definition: Almond-shaped structure in temporal lobe; central role in emotional processing, especially fear.

Functions: - Emotional memory formation - Fear conditioning and response - Processing of threatening stimuli - Social and emotional behavior

Who Overuses: Individuals with anxiety disorders show hyperactivity; overactivation in PTSD, phobias, social anxiety.

Source: LeDoux, J. (2007). The amygdala. *Current Biology*.

9. Hippocampus

Definition: Seahorse-shaped structure in medial temporal lobe; critical for memory formation.

Functions: - Consolidation of declarative/explicit memories - Spatial navigation and memory - Contextual memory

Neurogenesis: One of few brain areas with adult neurogenesis.

Damage: Anterograde amnesia (inability to form new memories) as seen in H.M. case.

Source: Squire, L. R., & Wixted, J. T. (2011). The cognitive neuroscience of human memory since H.M. *Annual Review of Neuroscience*.

10. Cerebellum

Definition: Structure at base of brain; traditionally known for motor coordination.

Functions: - Motor coordination and balance - Procedural learning (skills/habits) - Timing and sequencing - Cognitive functions (attention, language, fear conditioning) - "Little brain" - contains half the brain's neurons

Source: Koziol, L. F., et al. (2014). Consensus paper: The cerebellum's role in movement and cognition. *The Cerebellum*.

11. Limbic System

Definition: Set of brain structures involved in emotion, motivation, memory, and olfaction.

Key Components: - **Amygdala:** Fear, emotional processing - **Hippocampus:** Memory formation - **Hypothalamus:** Homeostasis, hormones, drives - **Cingulate cortex:** Emotional regulation, pain processing - **Septal nuclei:** Reward, pleasure

Source: MacLean, P. D. (1990). *The triune brain in evolution*. Plenum Press.

12. Reticular Formation

Definition: Network of neurons in brainstem; regulates arousal and sleep-wake transitions.

Functions: - Ascending reticular activating system (ARAS) - maintains consciousness - Sleep-wake cycle regulation - Habituation to repetitive stimuli - Muscle tone regulation - Cardiovascular and respiratory control

Source: Steriade, M. (1996). Arousal: Revisiting the reticular activating system. *Science*.

13. Pituitary Gland

Definition: "Master gland" at base of brain; produces hormones and regulates other endocrine glands.

Hormones Produced: - Growth hormone (GH) - Thyroid-stimulating hormone (TSH) - Adrenocorticotropic hormone (ACTH) - Prolactin - Gonadotropins (FSH, LH) - Antidiuretic hormone (ADH) - Oxytocin

Regulated by: Hypothalamus via releasing/inhibiting hormones.

Source: Pocock, G., & Richards, C. D. (2006). *Human physiology* (2nd ed.). Oxford University Press.

14. Synapse

Definition: Junction between two neurons (or neuron and effector cell) where signals are transmitted.

Components: - Presynaptic terminal (sending neuron) - Synaptic cleft (gap) - Postsynaptic membrane (receiving neuron) - Neurotransmitters

Process: Action potential → neurotransmitter release → binding to receptors → postsynaptic potential.

Types: Excitatory, inhibitory, modulatory.

Source: Kandel, E. R., et al. (2013). *Principles of neural science* (5th ed.). McGraw-Hill.

15. Dendrite

Definition: Branch-like extensions from neuron cell body that receive signals from other neurons.

Functions: - Receive neurotransmitter input from other neurons - Integration of synaptic signals - Contribute to neural computation

Dendritic Arborization: Extensive branching increases surface area for synaptic connections.

Development: Dendritic growth and pruning are key processes in brain development.

Source: Kandel, E. R., et al. (2013). *Principles of neural science* (5th ed.). McGraw-Hill.

16. Axon

Definition: Long, slender projection of neuron that conducts electrical impulses away from cell body.

Characteristics: - Can be myelinated (faster conduction) or unmyelinated - Action potentials travel "all-or-none" - Terminal branches form synapses with other neurons - Axon hillock: integration zone where action potential initiated

Myelination: Process of forming myelin sheath (oligodendrocytes in CNS, Schwann cells in PNS).

Source: Kandel, E. R., et al. (2013). *Principles of neural science* (5th ed.). McGraw-Hill.

17. Synaptic Pruning

Definition: Process by which unused neural connections (synapses) are eliminated during brain development.

Timeline: - Begins in infancy - Two major waves: early childhood (ages 3-6) and adolescence (beginning at puberty) - Adolescent pruning particularly affects prefrontal cortex - "Use it or lose it" principle

Significance: Essential for brain efficiency; over-pruning or under-pruning implicated in disorders like schizophrenia and autism.

Source: Huttenlocher, P. R. (1979). Synaptic density in human frontal cortex. *Brain Research*.

18. Terminal Branches (Axon Terminals)

Definition: Branching endings of an axon that form synapses with other neurons or effector cells.

Functions: - Release neurotransmitters into synaptic cleft - Contain synaptic vesicles with neurotransmitters - Site of signal transmission to next cell

Structure: Contains mitochondria, vesicles, and release machinery; separated from postsynaptic cell by synaptic cleft.

Source: Kandel, E. R., et al. (2013). *Principles of neural science* (5th ed.). McGraw-Hill.

19. Brain Plasticity

Definition: Brain's ability to change structure and function in response to experience, learning, or injury.

Types: - **Experience-Expectant Plasticity:** Normal development requires expected environmental input (e.g., visual experience) - **Experience-Dependent Plasticity:** Specific learning creates new synaptic connections

Key Periods: - Critical/sensitive periods: heightened plasticity - Throughout lifespan but decreases with age - Greater in childhood than adulthood

Applications: Recovery from stroke, learning, rehabilitation.

Source: Kolb, B., & Gibb, R. (2011). Brain plasticity and behaviour in the developing brain. *Journal of the Canadian Academy of Child and Adolescent Psychiatry*.

20. Mirror Neurons

Definition: Neurons that fire both when an individual performs an action and when observing another perform the same action.

Location: Primarily in premotor cortex and inferior parietal lobule.

Proposed Functions: - Action understanding - Imitation learning - Theory of mind - Empathy - Language development

Controversy: Some debate about human evidence and specific functions.

Source: Rizzolatti, G., & Craighero, L. (2004). The mirror-neuron system. *Annual Review of Neuroscience*.

21. Neurotransmitters Overview

Dopamine: Reward, motivation, motor control, pleasure. Associated with: Parkinson's (too little), Schizophrenia (too much in some pathways).

Serotonin: Mood, sleep, appetite, pain. Associated with: Depression (SSRIs increase serotonin).

Norepinephrine: Arousal, attention, fight-or-flight. Associated with: ADHD, anxiety.

GABA: Primary inhibitory neurotransmitter. Associated with: Anxiety disorders, epilepsy, sleep.

Glutamate: Primary excitatory neurotransmitter. Associated with: Learning, memory, excitotoxicity.

Acetylcholine: Muscle contraction, attention, memory. Associated with: Alzheimer's (depleted), Myasthenia gravis.

Source: Nestler, E. J., et al. (2015). *Molecular neuropharmacology: A foundation for clinical neuroscience* (3rd ed.). McGraw-Hill.

22. Prenatal Development

Germinal Period (0-2 weeks): Zygote forms, rapid cell division, implantation in uterine wall.

Embryonic Period (2-8 weeks): Major organ systems form; most vulnerable to teratogens; neural tube develops.

Fetal Period (9 weeks-birth): Rapid growth and refinement of organ systems; brain development accelerates; viability around 24 weeks.

Age of Viability: ~24 weeks gestation (earliest survival outside womb with intensive care).

Source: Moore, K. L., et al. (2019). *The developing human: Clinically oriented embryology* (11th ed.). Elsevier.

23. Teratogens

Definition: Environmental agents that can cause prenatal developmental abnormalities or death.

Examples: - **Alcohol:** Fetal Alcohol Spectrum Disorders (FAS, pFAS, ARND) - **Thalidomide:** Limb defects - **Mercury:** Neurological damage - **Radiation:** Microcephaly, intellectual disability - **Infections:** Rubella, Zika, CMV, Toxoplasmosis - **Drugs:** Cocaine, nicotine

Timing Matters: Effects depend on when exposure occurs during development.

Source: Moore, K. L., et al. (2019). *The developing human* (11th ed.). Elsevier.

24. Neurodevelopmental Processes

Neurogenesis: Production of new neurons (mostly prenatal).

Migration: Neurons move to final locations in brain.

Differentiation: Neurons develop specialized structures and functions.

Synaptogenesis: Formation of synaptic connections (rapid in first 2 years).

Myelination: Formation of myelin sheaths (continues into early adulthood, especially frontal lobes).

Synaptic Pruning: Elimination of excess synapses (peaks in early childhood and adolescence).

Source: Stiles, J., & Jernigan, T. L. (2010). The basics of brain development. *Neuropsychology Review*.

25. Neonatal Reflexes

Moro Reflex: Startle response to sudden loss of support or loud noise; arms extend then flex. Present at birth, disappears 4-6 months.

Palmar Grasp: Objects placed in palm trigger strong grip. Present at birth, fades 3-4 months.

Rooting: Touching cheek causes head turning toward stimulus. Present at birth, fades ~4 months.

Sucking: Rhythmic sucking when roof of mouth stimulated. Present at birth.

Babinski: Stroking sole causes toes to fan upward. Present at birth, disappears ~12 months.

Tonic Neck (Fencing): Head turned to side causes extension of arm/leg on that side. Fades ~6 months.

Clinical Significance: Absence/persistence can indicate neurological problems.

Source: Prechtl, H. F. R. (1977). *The neurological examination of the full-term newborn infant*.

26. Prenatal Genetic Terms

Zygote: Single cell formed when sperm fertilizes egg; contains 46 chromosomes (diploid). First 2 weeks.

Gametes: Reproductive cells (sperm/ova) with 23 chromosomes (haploid); formed through meiosis.

Genotype: Genetic makeup (DNA sequence).

Phenotype: Observable characteristics resulting from genotype + environment.

Allele: Different versions of a gene.

Polygenic Inheritance: Multiple genes contribute to phenotype (height, intelligence, skin color).

Epigenetics: Changes in gene expression not involving DNA sequence changes; influenced by environment.

Source: Plomin, R., et al. (2016). *Behavioral genetics* (7th ed.). Worth Publishers.

27. Infant States

Vernix: Waxy, cheese-like coating on fetal/newborn skin; protective, washes off shortly after birth.

Lanugo: Fine, soft hair covering fetus (especially 5th-6th month); typically shed before birth, may remain on premature infants.

Menarche: First menstruation; average age ~12-13 years (range 10-16).

Source: Moore, K. L., et al. (2019). *The developing human* (11th ed.). Elsevier.